Boolean polynomial obstructions to binary expansion of balanced Cayley complexes

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Abstract

The proposed coboundary expansion of random balanced Cayley complexes with binary coefficients and logarithmically many group-product layers fails in every fixed dimension $k \geq 2$. For $G = (\mathbb{F}_2)^d$ and every $A \subseteq G$, we prove

$$\dim_{\mathbb{F}_2} H^{k-1}(Y_{A,k}; \mathbb{F}_2) \geq \sum_{j=0}^{\min(k,d)} \binom{d}{j} - |A|.$$

The proof maps a Boolean polynomial vanishing on A to a cohomology class by writing the polynomial of a total group sum as a simplicial coboundary on the complete join. Consequently, for any fixed positive constant C, every subset of size $\lceil C \log D(G) \rceil$ has zero binary coboundary expansion for all sufficiently large d, where $D(G)$ is the sum of the degrees of its irreducible complex representations.

1 The complex and the obstruction

Coboundary expansion measures the number of violated parity constraints relative to the distance of a binary cochain from the coboundary space. Meshulam introduced balanced Cayley complexes and proved a real-coefficient spectral-gap estimate when the number of allowed group products is logarithmic in a representation parameter [1]. Conjecture 8.1 of that paper asks for a binary coboundary-expansion analogue, uniformly over finite groups whose orders tend to infinity. We give a deterministic obstruction on elementary abelian 2-groups.

Fix integers $d \geq 1$ and $k \geq 2$, and write $G = (\mathbb{F}_2)^d$ additively. Let $V_i = \{i\} \times G$ for $1 \leq i \leq k+1$, and put $X = V_1 * \cdots * V_{k+1}$. A face of this join contains at most one vertex from each part. For $A \subseteq G$, let $Y_{A,k}$ contain the complete $(k-1)$-skeleton of X and exactly the k-faces with labels $(g_1, \ldots, g_{k+1})$ satisfying

$$g_1 + \cdots + g_{k+1} \in A.$$

Addition here is the ordered group product of the original construction.

All cochains below have coefficients in $\mathbb{F}_2$. We use the increasing order of vertex parts to identify a face with its tuple of labels. The simplicial coboundary $\delta_r : C^r(Y; \mathbb{F}_2) \to C^{r+1}(Y; \mathbb{F}_2)$ adds the values on the codimension-one faces. Write

$$B^r(Y) = \operatorname{im} \delta_{r-1}, \qquad Z^r(Y) = \ker \delta_r, \qquad H^r(Y; \mathbb{F}_2) = Z^r(Y)/B^r(Y)$$

for $r \geq 1$. For a cochain ϕ, let $|\phi| = |\operatorname{supp} \phi|$ be its unnormalized support count. The expansion constant under consideration is

$$h_{k-1}(Y) = \min_{\phi \notin B^{k-1}(Y)} \frac{|\delta_{k-1}\phi|}{\min_{b \in B^{k-1}(Y)} |\phi + b|}. \tag{1}$$

A nonzero cohomology class has a representative with zero numerator and strictly positive denominator in (1). Thus nonvanishing of $H^{k-1}(Y;\mathbb{F}_2)$ forces $h_{k-1}(Y)=0$.

Theorem 1. *For every $d\ge 1$, $k\ge 2$ and $A\subseteq(\mathbb{F}_2)^d$, put*

$$N_k(d)=\sum_{j=0}^{\min(k,d)}\binom{d}{j}.$$

Then

$$\dim_{\mathbb{F}_2} H^{k-1}(Y_{A,k};\mathbb{F}_2)\ge N_k(d)-|A|. \tag{2}$$

In particular, $|A|<N_k(d)$ implies $h_{k-1}(Y_{A,k})=0$.

For a finite group G, write $D(G)=\sum_{\rho\in\widehat{G}}\dim\rho$, where $\widehat{G}$ denotes its irreducible complex representations, as in [1]. For the groups considered here, $D((\mathbb{F}_2)^d)=2^d$, as proved below, so $\log D(G)=d\log 2$.

The number $N_k(d)$ is the dimension of the Boolean polynomial functions of degree at most k on $(\mathbb{F}_2)^d$. Vanishing on A imposes at most $|A|$ linear constraints on these functions. The central point is that the remaining functions give distinct cohomology classes in the specified complex. Since $N_k(d)$ is at least quadratic in d for $k\ge 2$, this already rules out logarithmic sampling.

Corollary 2. *Fix $k\ge 2$ and $C,\varepsilon>0$. For $G_d=(\mathbb{F}_2)^d$, let A be uniform among the subsets of size $m_d=\lceil C\log D(G_d)\rceil$. For all sufficiently large d, this sample size is admissible and*

$$\Pr\{h_{k-1}(Y_{A,k})\ge\varepsilon\}=0.$$

Consequently, the constants requested in the binary expansion conjecture do not exist for any fixed $k\ge 2$.

2 Proofs

Proof of Theorem 1. Let $\mathcal{P}_k(d)$ be the space of Boolean polynomial functions

$$q(t)=\sum_{\substack{S\subseteq[d]\\|S|\le k}} c_S\prod_{r\in S}t_r,\qquad c_S\in\mathbb{F}_2,$$

where $[d]=\{1,\dots,d\}$ and the empty product is one. These monomials are independent as functions. Indeed, evaluation at the vector 1_T supported on T gives $q(1_T)=\sum_{S\subseteq T}c_S$. For a zero function, the empty set first gives $c_\varnothing=0$, and induction on $|T|\le k$ then gives every $c_T=0$. Therefore $\dim\mathcal{P}_k(d)=N_k(d)$.

Let K_A be the subspace of polynomials vanishing on A. The evaluation map $\mathcal{P}_k(d)\to\mathbb{F}_2^A$ has rank at most $|A|$, so

$$\dim K_A\ge N_k(d)-|A|. \tag{3}$$

We construct a linear injection from K_A into the cohomology in (2).

For each $q\in\mathcal{P}_k(d)$, expand $q(g_1+\cdots+g_{k+1})$. For a monomial indexed by S, the expansion is

$$\prod_{r\in S}\left(\sum_{j=1}^{k+1}(g_j)_r\right)=\sum_{\alpha:S\to[k+1]}\prod_{r\in S}(g_{\alpha(r)})_r. \tag{4}$$

Each term uses at most $|S|\le k$ of the $k+1$ vertex parts. Assign it to the smallest part index i outside $\operatorname{im}\alpha$; for the empty monomial assign it to $i=1$. Grouping the assigned terms, with their coefficients c_S, yields functions f_i^q independent of g_i such that

$$q(g_1+\cdots+g_{k+1})=\sum_{i=1}^{k+1}f_i^q(g_1,\dots,\widehat{g_i},\dots,g_{k+1}). \tag{5}$$

The assignment is fixed independently of q, so the functions depend linearly on its coefficients.

Every $(k-1)$-face of X misses exactly one vertex part. Define Tq on a face missing part i to have value f_i^q at that face's labels. Equation (5) is precisely the identity

$$\delta_X(Tq)(g_1,\dots,g_{k+1}) = q(g_1+\cdots+g_{k+1}). \tag{6}$$

If $q \in K_A$, the right-hand side vanishes on every top face of $Y_{A,k}$. Hence Tq is a cocycle there, and $q \mapsto [Tq]$ is a linear map into $H^{k-1}(Y_{A,k};\mathbb{F}_2)$.

To prove injectivity, suppose $Tq = \delta_{k-2}\psi$ in $Y_{A,k}$. The complexes $Y_{A,k}$ and X have identical faces and incidences through dimension $k-1$, so the same equality holds in X. Applying its next coboundary gives $\delta_X(Tq) = 0$: in δ^2, every codimension-two subface occurs twice, and $2 = 0$ in $\mathbb{F}_2$. Equation (6), evaluated at $(t, 0, \dots, 0)$, now gives $q(t) = 0$ for every $t \in G$. This is a valid top face of X because its vertex parts are disjoint, even when some labels coincide. Thus $q = 0$, proving injectivity. Together with (3), it proves (2).

If $|A| < N_k(d)$, choose a nonzero $q \in K_A$. Then Tq is a cocycle outside $B^{k-1}(Y_{A,k})$. Each $Tq + b$, with b in that coboundary space, is nonzero, so its support has at least one face. The denominator in (1) is thus a finite positive minimum, while the numerator is zero. All ratios are nonnegative, and the asserted zero expansion follows. □

Proof of Corollary 2. Every irreducible complex representation of G_d is one-dimensional. Indeed, its coordinate generators act by commuting involutions. For an involution M, the projectors $(I+M)/2$ and $(I-M)/2$ split into its two eigenspaces. The other generators preserve both spaces, and successive splitting gives simultaneous eigenspaces. Every line in a nonzero simultaneous eigenspace is invariant, so irreducibility forces dimension one. The 2^d independent choices of signs on the coordinate generators give exactly the characters $\chi_u(t) = (-1)^{u\cdot t}$ for $u \in (\mathbb{F}_2)^d$. Consequently $D(G_d) = 2^d$.

Choose $d_0 \geq 2$ so that $(d_0+1)/2 > C\log 2$. For every $d \geq d_0$, using natural logarithms,

$$\begin{aligned} m_d = \lceil Cd\log 2\rceil &\leq Cd\log 2 + 1 < 1 + \frac{d(d+1)}{2} \\ &= 1 + d + \binom{d}{2} \leq N_k(d) \leq 2^d. \end{aligned}$$

The sample size is therefore admissible. Theorem 1 gives $h_{k-1}(Y_{A,k}) = 0$ for every subset of that size, making the displayed probability zero for every $\varepsilon > 0$. Since $|G_d| = 2^d \to \infty$, this sequence violates the required guarantee for every proposed pair of constants and each fixed $k \geq 2$. A different fixed base of the logarithm only rescales C. □

AI-use statement. OpenAI Codex assisted with mathematical derivation, proof checking and drafting of this manuscript. The mathematical claim and the complete proof are presented here for independent review.

References

[1] R. Meshulam, *Random Balanced Cayley Complexes*, arXiv:2211.02085v1 (2022), Sections 1–2 and Conjecture 8.1. `https://arxiv.org/abs/2211.02085v1`.